

Part 17: Clustering

Here, we will focus on the general problem of **clustering**, the challenge of dividing objects into K groups such that similar objects are assigned to the same group.

Why Cluster?

Making predictions is a very natural problem; most people can understand why it is useful to study methods for supervised learning. Applications for unsupervised learning can be less obvious.

There can be distinct groups present in data sets. This grouping can provide insight into the structure of the data, and guide further analyses (in the same way that stocks in different “sectors” may be best modeled separately).

Clustering can also help in the identification of outliers.

The following examples are from finance. Note that I am not making any claims as to the legitimacy and/or accuracy of their analyses; I am just providing examples of ways in which clustering techniques could be used.

Pavlidis, et al. (2006): “Financial Forecasting through Unsupervised Clustering and Neural Networks,” *Operational Research*.

The authors use clustering to divide the predictor space into regions of greater homogeneity, and fit distinct (neural network) models in each of these.

“Although global approximation methods can be applied to model and forecast time series . . . , it is reasonable to expect that forecasting accuracy can be improved if regions of the input space exhibiting similar dynamics are identified and subsequently a local model is constructed for each of them.”

Das (2003): “Hedge Fund Classification using K-means Clustering Method,” *9th International Conference on Computing in Economics and Finance*.

The author proposes using clustering as a means of categorizing hedge funds, hence improving on the existing “myriad of classifications, some overlapping and some mutually exclusive.”

Craighead and Klemesrud (2002): “Stock Selection Based on Cluster and Outlier Analysis,” Nationwide Financial, Columbus, OH.

The authors use clustering to identify outliers that could potentially create problems with portfolio selection algorithms.

Financial report		d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8	d_9
Qualitative features	efficient	1	1	0	1	0	0	1	0	0
	growth	1	1	0	0	0	0	0	0	0
	advantage	1	1	1	0	0	0	0	0	0
	improvement	1	0	0	0	0	0	0	0	0
	deficient	0	0	0	1	1	0	0	0	0
	reorganize	0	0	0	1	1	0	0	1	1
	difficulty	0	0	0	1	1	1	1	0	0
	complaint	0	0	0	1	0	0	0	0	1
Quantitative features	operating margin	0.4	0.38	0.37	0.1	0.07	0.08	0.05	0.06	0.03
	ROE	0.3	0.28	0.27	0.01	0.04	0.04	0.07	0.02	0.05
	ROTA	0.25	0.23	0.22	0.02	0.05	0.07	0.04	0.01	0.04
	equity to capital	0.8	0.78	0.77	0.45	0.4	0.5	0.45	0.5	0.55
	receivables turnover	2.5	2.4	2.45	1.4	1.3	1.5	1.2	1.1	1.5
Class label		1	1	0	-1	-1	-1	0	-1	-1

Table 1. An example dataset.

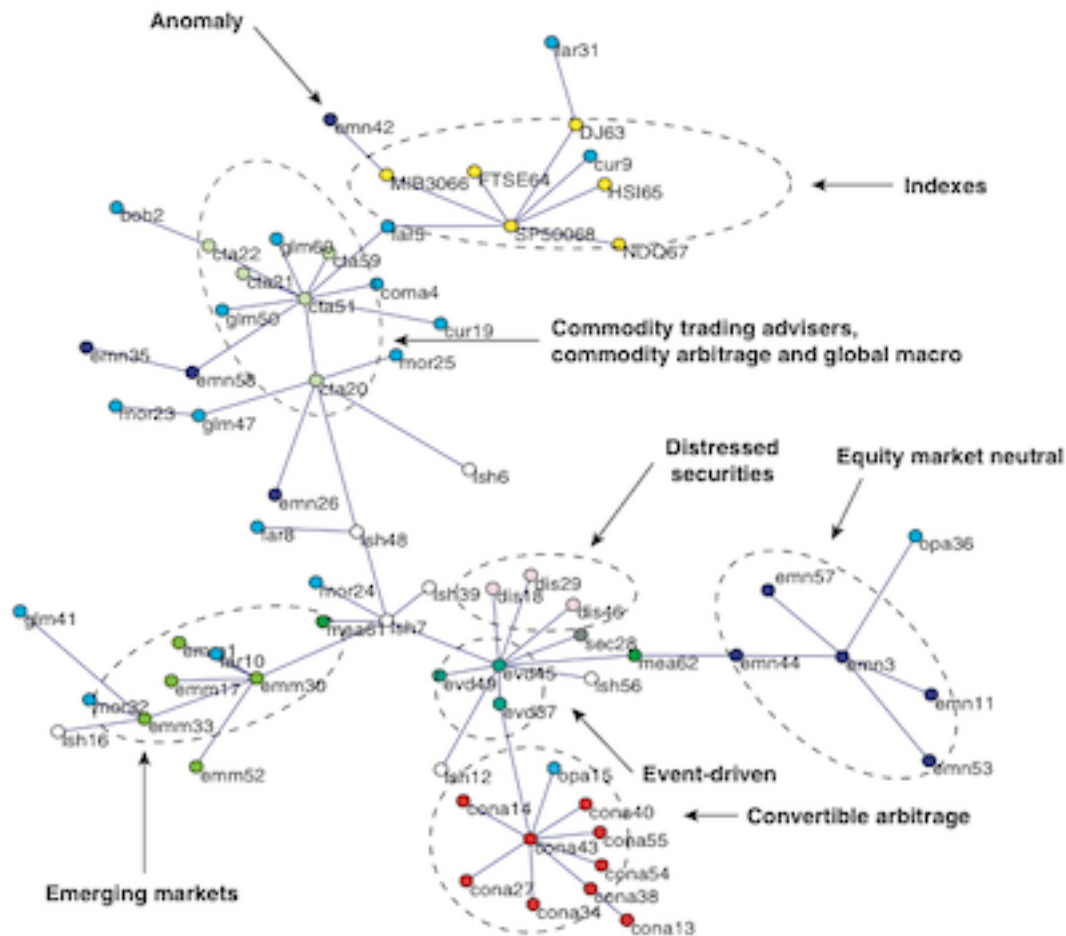
From Lee, et al. (2010)

Lee, et al. (2010): “An Effective Clustering Approach to Stock Market Prediction,” *PACIS 2010 Proceedings*

The authors use clustering to divide the financial reports into homogeneous groups, and then take the cluster center as a “representative.”

“When a financial report is released, we will transform it into a feature vector ... [and] we assign [it] to the nearest representative feature vector. Then, we predict the direction of the stock price movement according to the class label of the nearest representative feature vector.”

In this framework, the clustering serves as a “smoothing” operation on “nonstandard” data (the financial report).



From Miceli and Susinno (2003)

Miceli and Susinno (2003): "Using Trees to Grow Money," *Risk*

The authors cluster hedge fund performance time series in order to "visualize a taxonomy embedded in synchronous historical data." They state that "[i]n a universe with low transparency from an investor's point of view, and where operating strategies are protected from full disclosure . . . instruments of classification . . . would allow the investor to verify the declared strategies in the statements issued by hedge funds. In particular, nodes anomalous to a reference cluster help to identify funds that require a deeper investigation."

Leon, et al. (2017): “Clustering algorithms for Risk-Adjusted Portfolio Construction”

“This paper aims at placing stocks into groups, or clusters, suggested by the raw data, not defined a priori, such that stocks in a given cluster tend to be similar to each other and stocks in different clusters tend to be dissimilar. In terms of financial theory, this purpose allows building highly diversified portfolios. In the following stage, this work proposes a risk-adjusted optimization using Sharpe Ratio for optimal asset allocation into each cluster.”

K-Means Clustering

The **K-means clustering algorithm** is a relatively simple and intuitive approach to dividing the sample $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ into K distinct groups.

Despite repeated extensions and enhancements of the method and the growth of other clustering methods, K-means remains popular and useful.

The algorithm is based on a very natural measure of within-cluster heterogeneity. Observations are assigned to clusters in an effort to minimize this measure.

It is assumed that K is fixed by the user; the choice of K is a tricky issue.

I will adopt the notation used by *Introduction to Statistical Learning*, Section 10.3.1.

First, define C_k to be the set of indices that belong to cluster k , for $k = 1, 2, \dots, K$.

Each observation is assigned to exactly one cluster. In other words, $C_k \subseteq \{1, 2, \dots, n\}$ with

$$\bigcup_{k=1}^K C_k = \{1, 2, \dots, n\} \text{ and } C_k, C_\ell \text{ disjoint for } k \neq \ell.$$

With K-means, the similarity within cluster k is measured by

$$W(C_k) = \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{kj})^2$$

where $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $\bar{\mathbf{x}}_k$ is the cluster **centroid**, calculated as the sample mean of the cluster members.

Exercise: Under what circumstance(s) will $W(C_k)$ be small?

K-means clustering seeks to find the allocation of the observations that minimizes the **inertia**:

$$\sum_{k=1}^K W(C_k)$$

Comparing every possible allocation is not realistic. (Note that there is no restriction on the sizes of the clusters.)

Instead, the following algorithm is utilized: (Algorithm 10.1 in ISL.)

Randomly assign a number from 1 to K to each of the observations. These are the initial cluster assignments. Then repeat until no changes are made to cluster assignments:

1. Find the centroid for each of the K clusters.
2. Re-assign each observation to the cluster whose centroid is closest as measured by Euclidean distance.

Comment: Note that the criterion is guaranteed to not increase with each iteration of the algorithm, but the search could get caught in a local minimum (and not the global minimum).

To address this problem, it is standard to repeat the algorithm several times with different (randomly chosen) starting allocations. If you keep track of the criterion achieved from each repetition, you can easily determine which allocation is the best.

K-means in Python

K-means is implemented in Python using the function `KMeans()` that is part of `scikit-learn`.

We will cluster our data set of random-chosen NYSE stocks, with data taken from August 18, 2017 to September 29, 2017. This is a total of 30 trading days.

```
In [1]: import pandas as pd
import numpy as np

stocksample = pd.read_csv("stocksample.txt",
    sep="\t")

stocksample['sector'].fillna("None",
    inplace=True)

stocksamplescl = stocksample.iloc[:,4:34]

for i in range(1000):
    stocksamplescl.iloc[i,:] = \
        (stocksamplescl.iloc[i,:] - \
         np.mean(stocksamplescl.iloc[i,:]))/\
         np.std(stocksamplescl.iloc[i,:])
```

The syntax is simple. The function `KMeans()` initializes the process, while the `fit()` function finds the clusters.

```
In [2]: from sklearn.cluster import KMeans

        kminit = KMeans(n_clusters=5, n_init=10)
        kmout = kminit.fit(stocksamplescl)
```

Here I have specified that I want $K = 5$ clusters, and that I want the basic algorithm repeated ten times in order to increase the chances of finding the global minimum.

Some comments on the output:

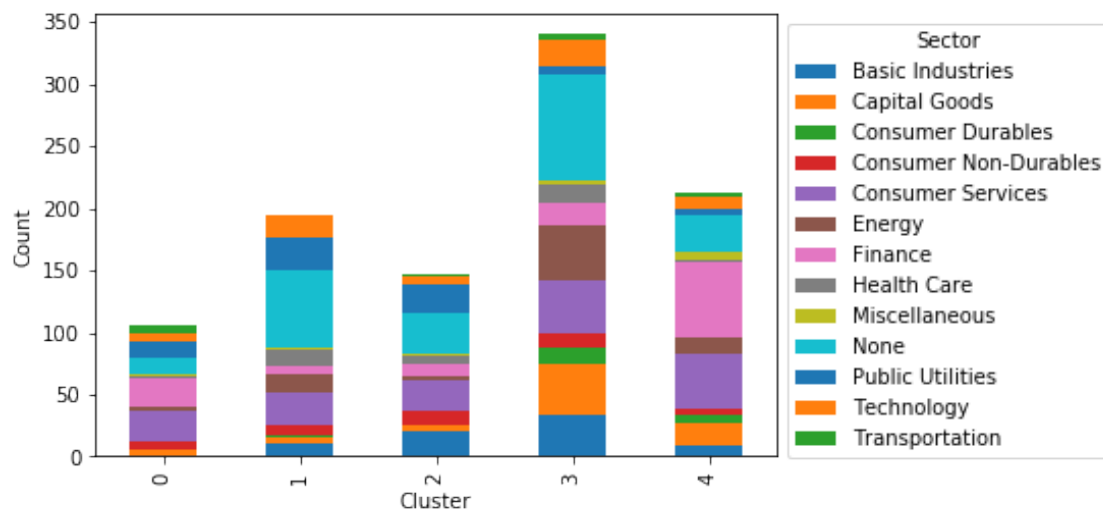
1. The rows of `kmout.cluster_centers_` holds the centers of the K clusters. In our example, each center is a vector of length 30.
2. `kmout.labels_` specifies the cluster that each of the observations has been assigned to.
3. The value of the inertia can be found from `kmout.inertia_`.

We can inspect how the clusters vary with respect to sector.

```
In [3]: stocksample["kmclust"] = kmout.labels_
stocksample["kmclust"] = \
    stocksample["kmclust"].\
    astype('category').cat.as_unordered()

import matplotlib.pyplot as plt

df = stocksample.groupby(\
    ["kmclust", "sector"])["symb"].\
    count().unstack("sector").fillna(0)
df.plot(kind='bar', stacked=True,
        legend=False)
plt.xlabel("Cluster")
plt.ylabel("Count")
plt.legend(title="Sector", bbox_to_anchor=(1.0, 1.0))
plt.show()
```



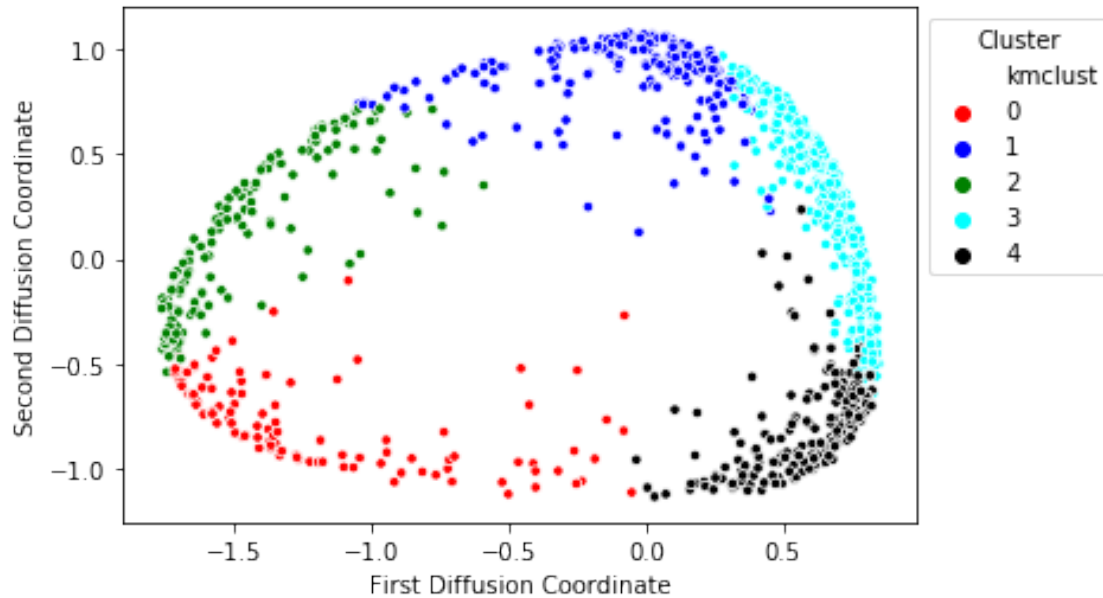
It is also interesting to compare K-means with the output of a dimension reduction technique seen previously:

```
In [4]: from pydiffmap import diffusion_map as dm
import seaborn as sns

dmap = dm.DiffusionMap.from_sklearn(n_evecs=3,
    epsilon=50)
dmapoutstocks = \
    dmap.fit_transform(stocksamplescl)

stocksampledmap = pd.concat([stocksample,
    pd.DataFrame(dmapoutstocks, columns=\
        ["dmap1", "dmap2", "dmap3"])], axis=1)

ax = sns.scatterplot(x="dmap1", y="dmap2",
    data=stocksampledmap,
    s=20, hue="kmclust",
    palette=['red', 'blue', 'green', 'cyan', "black"])
plt.xlabel("First Diffusion Coordinate")
plt.ylabel("Second Diffusion Coordinate")
plt.legend(title="Cluster",
    bbox_to_anchor=(1.0, 1.0))
plt.show()
```



Exercise: Interpret the plot above.

Exercise: Would it seem sensible to run K-means directly on the low-dimensional representation, e.g., on the diffusion coordinates?

Hierarchical Clustering

Another class of clustering procedures, called **hierarchical clustering**, are characterized by the distinctive **dendrograms** they create to depict the relationships between observations.

An appealing feature of these methods is that the creation of the dendrogram does not require the specification of the number of clusters. Sometimes, the dendrogram may reveal an “obvious” division among the observations, and hence suggest the number of clusters needed.

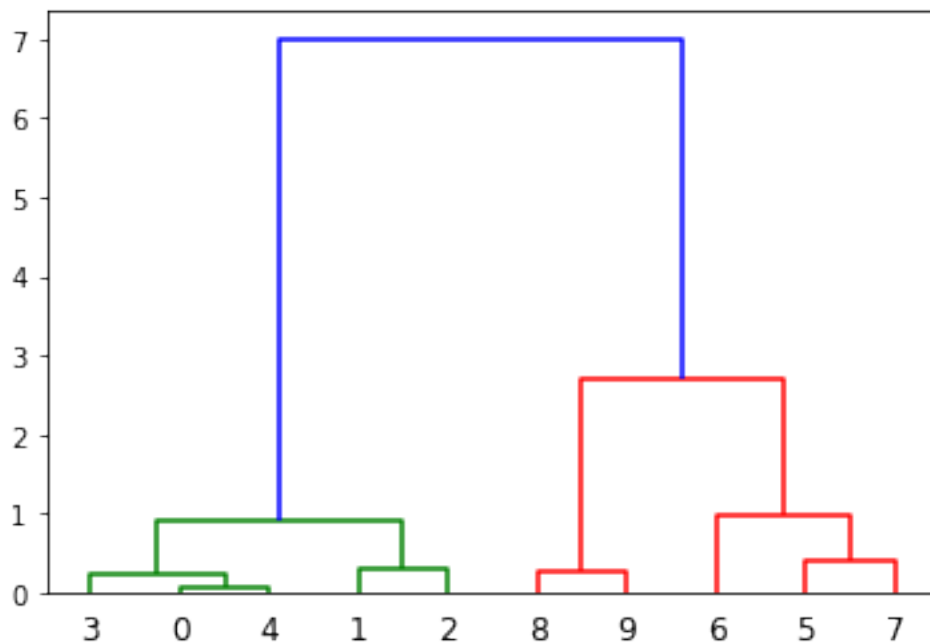
This clustering method allows the user to specify a measure of dissimilarity. Often, Euclidean distance is used (just as in K-means).

It is useful to begin by understanding how to interpret the dendrogram. An example is given below.

```
In [5]: np.random.seed(0)
        x = np.random.normal(\
            loc=[0,0,0,0,0,5,5,5,7,7],
            scale=0.5, size=10)

        from scipy.cluster import hierarchy

        Z = hierarchy.linkage(x.reshape(10,1),
                               "complete")
        hierarchy.dendrogram(Z)
None
```



Observations which are similar are “linked” at a low height in the dendrogram. For instance, to link observations 5 and 6, you need only go up to a height of approximately 1.0. These two observations are quite similar. But, in order to link observations 1 and 9, you need to go to a height of approximately 7. These are quite dissimilar.

The units for “height” match the units of the dissimilarity metric that the user provides.

Exercise: Which pair are more similar: 5 and 7, or 3 and 4?

Clusters can be formed by “cutting” the dendrogram at a user-chosen height. It may seem “clear” by looking at the dendrogram that cutting at around a height of 3 would be a good choice. The results is three clusters.

Note that in this case I generated the ten observations from the normal distribution with a standard deviation of 0.5. The mean for observations 0 thru 4 was 0, for observations 5 thru 7 was 5, and for observations 8 and 9 was 7. This explains the structure in the dendrogram.

Building the Dendrogram

The process of constructing the dendrogram is quite intuitive.

First, the two observations which are the most similar (as measured by the chosen dissimilarity metric) and joined together, at a height equal to their dissimilarity. In our example above, observations 0 and 4 were the first to be joined, at a height very close to zero.

This process then continues, with the most similar objects joined together at the appropriate height. **But:** Once observations are joined, they are **treated as a unit** from that point on. For instance, in the second step in the construction of the dendrogram above, observation 3 was joined with the **unit consisting of 0 and 4**.

This process continues until all observations are joined.

The obvious question is thus: **How is the dissimilarity of collections of observations calculated?** There are a few different approaches. Two of the standard ones are as follows:

With **complete linkage**, the distance between sets of observations, S_i and S_j , is defined to be the **maximal** dissimilarity between all pairs $x \in S_i$ and $y \in S_j$.

With **single linkage**, the distance between sets of observations, S_i and S_j , is defined to be the **minimal** dissimilarity between all pairs $x \in S_i$ and $y \in S_j$.

Less-commonly used options are **centroid linkage**, in which the distance between S_i and S_j is calculated as the dissimilarity between the centroids of the two groups, and **average linkage**, calculated as the average distance between all pairs of elements in S_i and S_j .

Hierarchical Clustering in Python

The function `hierarchy()` from `scipy.cluster` performs hierarchical clustering in Python.

This function, by default, calculates the Euclidean distance between the provided **rows** of the data in order to construct the clusters.

The syntax for complete linkage clustering is as follows. (Note that only the first 100 stocks are used, since the dendrogram becomes difficult to read otherwise. There is no such practical limit in using this method, however.)

Other choices for `method` include `single`, `centroid`, and `average`.

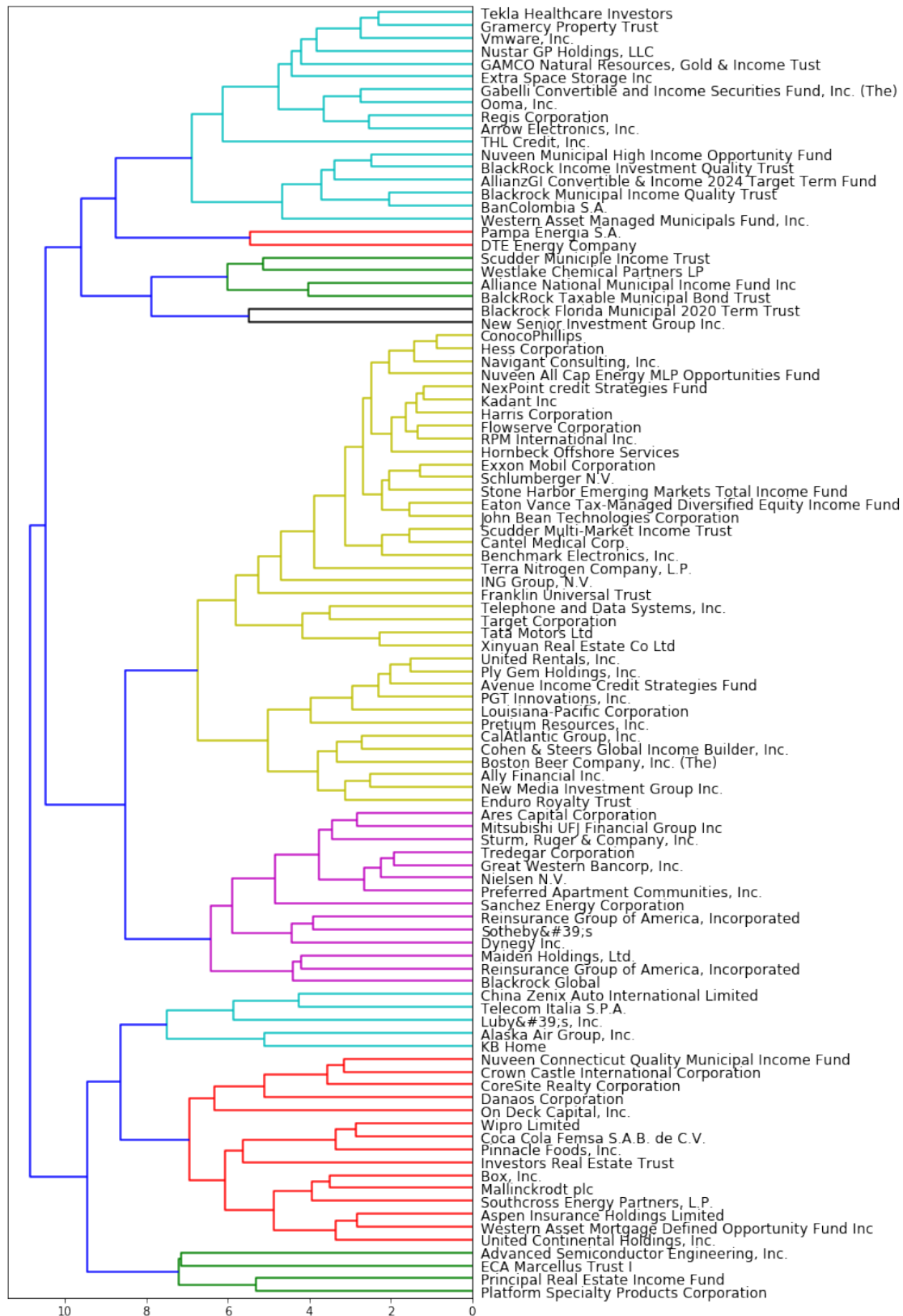
```
In [6]: hcout = hierarchy.linkage(\
        stocksamplescl[:100], method="complete")
```

The function `dendrogram()` constructs the dendrogram.

```
In [7]: fig, axs = plt.subplots(1, 1,
        figsize=[7,20])

        labs = pd.array(stocksamples.name\
        [0:100]).astype("str")
        hierarchy.dendrogram(hcout, ax=axs,
        orientation="left", labels=labs,
        leaf_font_size=12)

None
```



Another useful function is `fcluster()`. This will return cluster membership for each of the observations. With `criterion = "maxclust"`, the argument `t` specifies the number of clusters to return.

```
In [8]: hierarchy.fcluster(hcout, t=5,  
                           criterion = "maxclust")[0:5]
```

```
Out[8]: array([3, 2, 5, 2, 2], dtype=int32)
```

With `criterion = "distance"`, the argument `t` specifies the level at which to cut the tree.

```
In [9]: hierarchy.fcluster(hcout, t=10,  
                           criterion = "distance")[0:5]
```

```
Out[9]: array([2, 1, 3, 1, 1], dtype=int32)
```

We can again inspect how the clusters vary with respect to sector:

```
In [10]: hcout = hierarchy.linkage(stocksamplescl,
    "complete")
stocksampl["hcclust"] = hierarchy.fcluster\
    (hcout, t=10, criterion="maxclust")
stocksampl["hcclust"] = stocksampl["hcclust"].\
    astype('category').cat.as_unordered()

df = stocksampl.groupby(["hcclust", "sector"])\
    ["symb"].count().unstack("sector").fillna(0)
df.plot(kind='bar', stacked=True, legend=False)
plt.xlabel("Cluster")
plt.ylabel("Count")
plt.legend(title="Sector", bbox_to_anchor=(1.0, 1.0))
plt.show()
```

